

BBL 133 Mass and Energy Balance

CHAPTER 2

Energy Balances on Nonreactive Processes

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Open System

- **Energy Balance for an Open System**

$$\Delta \dot{H} + \Delta \dot{E}_k + \Delta \dot{E}_p = \dot{W}_s + \dot{Q}$$

Open system in which shaft-work and KE and PE changes can be neglected, energy balance reduces to

$$\dot{Q} = \Delta \dot{H}$$

$$\dot{Q} = \sum_{\text{out}} \dot{n}_i \hat{H}_i - \sum_{\text{in}} \dot{n}_i \hat{H}_i$$

Where, $\hat{H}_i$ - Specific enthalpies at their respective process conditions (such as T, P, physical state) relative to those components at some reference conditions

Closed System

For closed constant volume system ($W = 0$), energy balance is

$$Q = \Delta U$$

$$Q = \sum_{\text{final}} n_i \hat{U}_i - \sum_{\text{initial}} n_i \hat{U}_i$$

Here, $\hat{U}_i$ - Specific Internal Energies

- For some species, specific enthalpies and internal energies at specified conditions could be found in tables, but such data couldn't be found for every species.
- When table of $\hat{H}$ and $\hat{U}$ are not available for all process species, the procedure of evaluating $\Delta \dot{H}$ and ΔU is presented in this chapter.

Objective

- **Calculation of $\Delta\hat{H}$ and $\Delta\hat{U}$ for changes in state of species**
 - Isothermal changes in pressure
 - Isobaric changes (constant pressure) in temperature
 - Isothermal isobaric phase changes or mixing of two or more species
- **For a given reference state (T, P, Phase) and a process state for a species**
 - a) Calculation of $\hat{U}$ and $\hat{H}$ for the species at the process relative to the species at the reference state: Choose a path from the reference state to the process
 - b) Calculation of heat transfer or heat transfer rate for any given non-reactive process: Write mass and energy balance and solve them simultaneously for requested quantities

Objective

○ **Psychrometric chart or Humidity chart**

- Dry-bulb temperature, wet-bulb temperature, and humid volume of humid air
- For given value of any two of the variables plotted on the psychrometric chart (dry bulb and wet bulb temp, abs and relative humidity, dew point, humid volume), determine the remaining variable values and sp. enthalpy of the humid air
- You need to carry out material and energy balance calculations on heating, cooling, humidification or dehumidification process

○ **Adiabatic cooling process**

- Working of spray cooling & humidification, spray dehumidification and spray drying?
- How it is possible to dehumidify air by spraying water into it?

Reference State for $\hat{U}$ and $\hat{H}$

$\hat{U}$ (kJ/mol) = Specific IE = Sum of energies of motion of all N_A number of molecules in 1 g mole of species + Intramolecular KE and PE of all atoms and subatomic particles (Quantities difficult to determine)

$\hat{H} = \hat{U} + P\hat{V} \Rightarrow$ If we can't know the value of $\hat{U}$, we can't know $\hat{H}$ at specified state

- The absolute value of $\hat{U}$ or $\hat{H}$ at specified state can never be known
- We can only determine $\Delta\hat{U}$ and $\Delta\hat{H}$ for specified changes of state
- To determine $\Delta\hat{U}$ for change of state (or transition) from the reference state to other states, one has to arbitrary choose a reference state of a species
- If $\hat{U}_{\text{ref}} = 0$, then $\Delta\hat{U} = \hat{U}$ for specified state and is called specific IE at that state relative to reference state

Estimation of $\hat{U}$ and $\hat{H}$ using steam table

- Choose reference state = Liquid water at triple point
i. e., H_2O (l, 0.01°C , 0.00611 bar) at which $\hat{U} = 0$

- For water vapor @ 400°C and 10 bar: $\hat{U} = 2958$ kJ/kg
- This is not the absolute value of $\hat{U}$ for water in the specified state as we can't know the absolute value of $\hat{U}$
- This is specific IE of water vapor at 400°C and 10 bar relative to water at reference state

H_2O (l, 0.01°C , 0.00611 bar) $\rightarrow$ H_2O (v, 400°C , 10 bar)

$\Delta\hat{U} = 2958$ kJ/kg (Sp. IE of water vapor relative to water at reference state)

Estimation of $\hat{U}$ and $\hat{H}$ using steam table

- Specific enthalpy of water @ 400°C and 10 bar, relative to water at same reference state

$$\hat{H} = \hat{U} + P\hat{V}$$

$$= 2958 \text{ kJ/kg} + 10 \text{ bar} \times 0.307 \text{ m}^3/\text{kg} \times 1000 \text{ L/m}^3 \times \frac{8.314 \times 10^{-3} \frac{\text{kJ}}{\text{mol-K}}}{0.08314 \left(\frac{\text{L-bar}}{\text{mol-K}} \right)}$$

$$= 3264 \text{ kJ/kg}$$

State Properties

- $\hat{U}$ and $\hat{H}$ are state properties, their values depends only on the state of the species – primarily on its temperature and state of aggregation (solid, liquid, or gas) and to a lesser extent on its pressure
- State property doesn't depend on how the species reaches its state
- Value of ΔU and ΔH (for closed system) and $\Delta \dot{H}$ (for open system) calculated for a process are independent of reference state chosen for calculations of $\hat{U}_i$ and $\hat{H}_i$

Hypothetical Process Paths

- **Calculation of IE and enthalpy changes are associated with these processes**

1. Change in P at **constant T and state of aggregation**
2. Change in T at **constant P and state of aggregation**
3. Phase changes **at constant T and P**
4. Mixing of two liquids **at constant T and P**
5. Dissociation of a gas or a solid in a liquid **at constant T and P**
6. Chemical reaction **at constant T and P**

Construction of hypothetical “process path” from the initial state to the final state will consist of a series of steps of given 6 types

Hypothetical Process Paths

- Calculate $\Delta\hat{H}$ for each step, add ΔH 's for the steps to calculate $\Delta\hat{H}$ for the total process
- Since $\hat{H}$ is a state property, $\Delta\hat{H}$ calculated for the hypothetical process path (constructed for convenience) is same as $\Delta\hat{H}$ for the path actually followed by the process.
- Similar procedure can be followed to calculate $\Delta\hat{U}$ for any process

Hypothetical Process Paths

Example: Calculate $\Delta\hat{H}$ for conversion of solid phenol at 25°C and 1 atm to phenol vapor at 300°C and 3 atm

Method 1- If a table of enthalpies for phenol is given, subtract $\hat{H}_{\text{initial}}$ from $\hat{H}_{\text{final}}$ state

$$\Delta\hat{H} = \hat{H} (\text{vapor, 300°C, 3 atm}) - \hat{H} (\text{solid, 25°C, 1 atm})$$

Method 2- If table is not given, construct a hypothetical process path from solid at 25°C and 1 atm to vapor at 300°C and 3 atm.

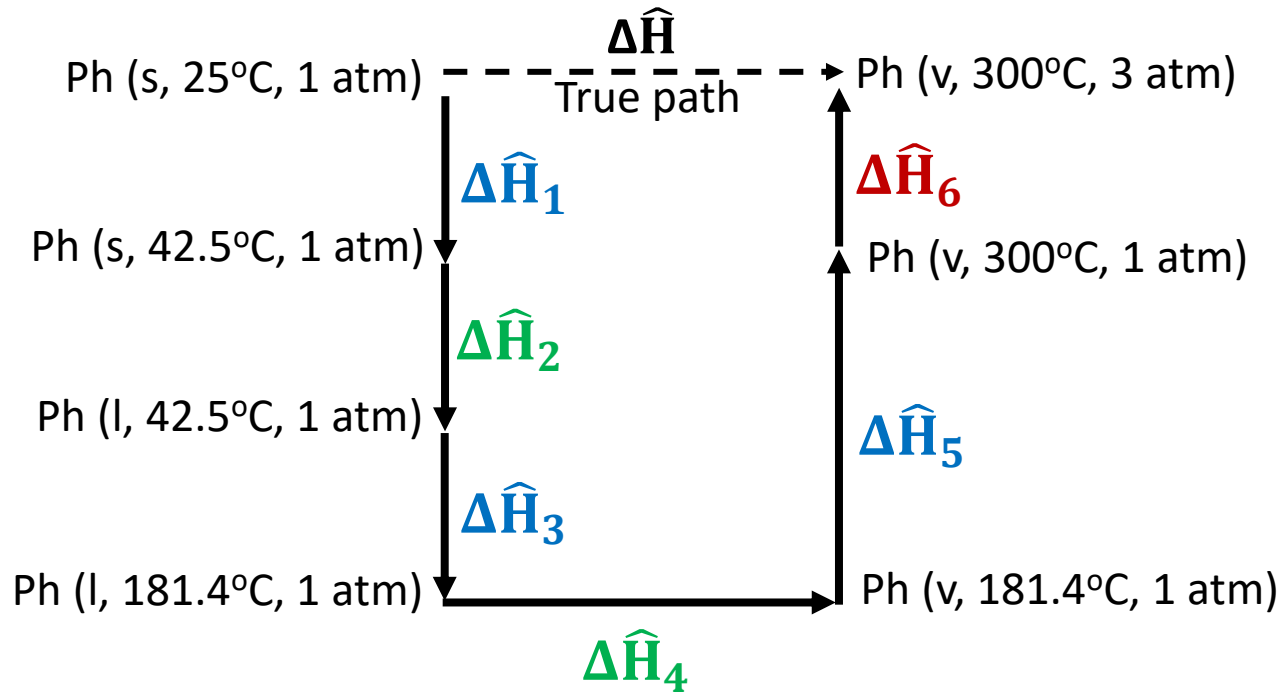
From physical property data,

$\Delta\hat{H}$ for melting of phenol: $\Delta\hat{H}_m = 42.5 \text{ kJ/mol @ } 42.5^\circ\text{C}$ (melting point) and 1 atm

$\Delta\hat{H}$ for vaporization of phenol: $\Delta\hat{H}_v = 25.77 \text{ kJ/mol @ } 181.4^\circ\text{C}$ (boiling point) and 1 atm

Hypothetical Process Paths

HYPOTHETICAL PROCESS PATH



Step 1, 3, and 5 : Change in T @ constant P and state

Step 2 and 4 : Change in phase (state) @ constant T and P

Step 6 : Change in P @ constant T and state

Desired:

$$\Delta \hat{H} = \Delta \hat{H}_1 + \Delta \hat{H}_2 + \Delta \hat{H}_3 + \Delta \hat{H}_4 + \Delta \hat{H}_5 + \Delta \hat{H}_6$$

Phase changes (step 2 and 4) occur at conditions for which tabulated enthalpy changes are available

Energy Balance Calculation Procedure

- In material balance, for given values of some process variables, values of other process variables have been calculated
- In energy balance calculations, calculation of heat transferred to or from the process system require writing and solving energy balance

Energy balance calculation procedure

- 1) Perform all required material balance calculations
- 2) Write energy balance (for closed or open system) and delete terms that are either zero or negligible for the given process system
- 3) Choose a reference state - (T, P, phase) – for each species involved in the process

Energy Balance Calculation Procedure

- 4) For closed constant volume system, construct “INLET-OUTLET Enthalpy Table” with columns for initial and final amount of each species (m_i or n_i) and specific internal energies relative to chosen reference state ($\hat{U}_i$).

Similarly, for an open system, construct table with columns for inlet and outlet stream component flow rates ($\dot{m}_i$ or $\dot{n}_i$) and specific enthalpies relative to chosen reference state ($\hat{H}_i$)

- 5) Calculate all required values of $\hat{U}_i$ (or $\hat{H}_i$) and insert values in appropriate places in the table: To perform calculation for a species in a particular state (inlet or outlet), choose any convenient path from the reference state to process state and determine $\hat{U}_i$ ($\hat{H}_i$) as $\Delta\hat{U}$ ($\Delta\hat{H}$) for that path

Energy Balance Calculation Procedure

6) Calculate

For closed system-

$$\Delta U = \sum_{\text{final}} n_i \hat{U}_i - \sum_{\text{initial}} n_i \hat{U}_i$$
$$\Delta U = \sum_{\text{final}} \dot{m}_i \hat{U}_i - \sum_{\text{initial}} \dot{m}_i \hat{U}_i$$

For open system-

$$\Delta \dot{H} = \sum_{\text{out}} \dot{n}_i \hat{H}_i - \sum_{\text{in}} \dot{n}_i \hat{H}_i$$
$$\Delta \dot{H} = \sum_{\text{out}} \dot{m}_i \hat{H}_i - \sum_{\text{in}} \dot{m}_i \hat{H}_i$$

7) Calculate any work, KE, or PE term that you have not dropped from the energy balance

8) Solve energy balance for whichever variable is unknown

- Closed system: $Q + W = \Delta U + \Delta E_k + \Delta E_p$

- Open system: $\dot{W}_s + \dot{Q} = \Delta \dot{H} + \Delta \dot{E}_k + \Delta \dot{E}_p$

Problem 1

A fuel gas containing 95 mole% methane and the balance ethane is burned completely with 25% excess air. The stack gas leaves the furnace at 900 deg C and is cooled to 450 deg C in a waste heat boiler, a heat exchanger in which heat lost by cooling gases is used to produce steam from liquid water for heating, power generation, or process applications.

- a) Taking as a basis of calculation 100 mol of the fuel gas fed to the furnace, calculate the amount of heat (kJ) that must be transferred from the gas in the waste heat boiler to accomplish the indicated cooling.
- b) Assume all the heat transferred from the gas goes into the steam production. How much saturated steam at 50 bar can be produced from boiler feed water at 40 deg C for the same basis of calculation?
- c) At what rate (kmol/s) must fuel gas be burned to produce 1250 kg steam per hour in the waste heat boiler? What is the volumetric flow rate(m³/s) of the gas leaving the boiler?

Problem 2

An adiabatic membrane separation unit is used to dry (remove water vapor from) a gas mixture containing 10.0 mole% $\text{H}_2\text{O}(\text{v})$, 10.0 mole% CO , and the balance CO_2 . The gas enters the unit at 30 deg C and flows past a semipermeable membrane. Water vapor permeates through the membrane into an air stream. The dried gas leaves the separator at 30 deg C containing 2.0 mole% $\text{H}_2\text{O}(\text{v})$ and the balance CO and CO_2 . Air enters the separator at 50 deg C with an absolute humidity of 0.002 kg $\text{H}_2\text{O}/\text{kg}$ dry air and leaves at 48 deg C. Negligible quantities of CO , CO_2 , O_2 , and N_2 permeate through the membrane. All gas streams are at approximately 1 atm.

- i) Draw and label a flow chart of the process and carry out a degree-of-freedom analysis to verify that you can determine all unknown quantities on the chart
- ii) Calculate the ratio of entering air to entering gas (kg humid air/mol gas)
- (iii) Calculate the mole fraction of water

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Change in Pressure at Constant Temperature

- **For all liquids and gases**, internal energy ($\hat{U}$) and specific volume ($\hat{V}$) are independent of pressure at constant temperature

If pressure of liquid or solid changes at constant T,

incompressible

$$\Delta\hat{V} = \Delta\hat{U} = 0$$

$$\Delta\hat{H} = \Delta\hat{U} + \Delta(P\hat{V}) \approx \hat{V}\Delta P$$

$$\Delta\hat{H} = \hat{V}\Delta P$$

megaWatt

Unit conversion: $\hat{V}\Delta P = 1.013 \times 10^{-4} \frac{\text{MW}}{\text{SG}} \Delta P$

specific Gravity

Change in Pressure at Constant Temperature

- **For an ideal gas**, both $\hat{H}$ and $\hat{U}$ are independent of pressure at constant temperature

For a gas undergoing isothermal pressure change,

$$\left. \begin{array}{l} \Delta \dot{U} \approx 0 \\ \Delta \dot{H} \approx 0 \end{array} \right\} \begin{array}{l} \text{- For gases, above } 0^\circ\text{C and below 1 atm} \\ \text{- No need to take this assumption} \\ \text{if } \hat{U}(P, T) \text{ or } \hat{H}(P, T) \text{ are given} \end{array}$$

- For non-ideal gas (far from ideal) or if they undergo large pressure changes:
 - > Use table of thermodynamic properties (such as steam table for water)
 - > or, thermodynamic correlation (beyond the scope of this course)

Change in Pressure at Constant Temperature

For changes in pressure at constant temperature:

$$\Delta \hat{U} = 0 \text{ (ideal gases)}$$

$$\approx 0 \text{ (solids, liquids, gases close to ideal)}$$

$$\Delta \hat{H} = 0 \text{ (ideal gases)}$$

$$\approx 0 \text{ (gases close to ideal)}$$

$$\approx \hat{V} \Delta P \text{ (solids and liquids)}$$

Changes in Temperature

Sensible heat and Heat capacities

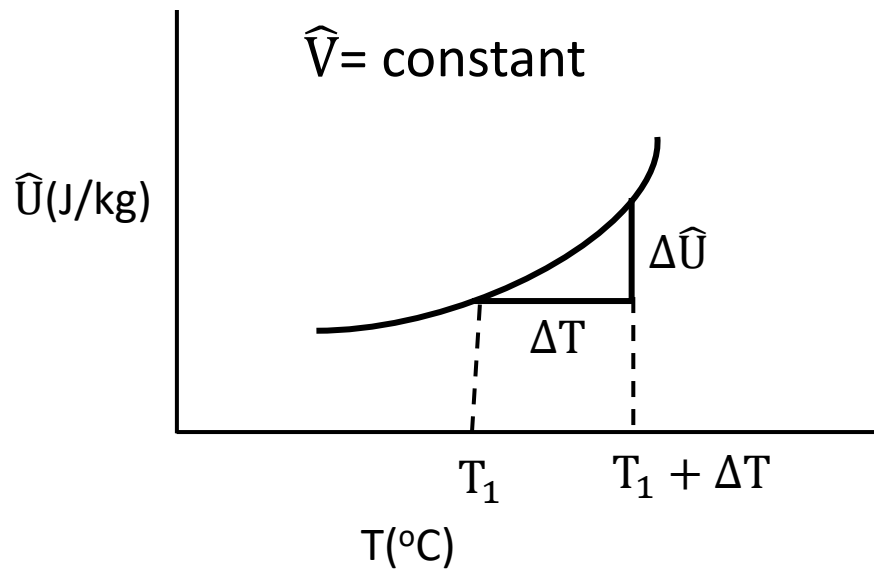
- Heat transferred to raise or lower the temperature of a substance or mixture of substances
- Quantity of heat required to produce a temperature change in a system can be determined from 1st law of thermodynamics

$$\left. \begin{array}{l} Q = \Delta U \text{ (Closed system)} \\ Q = \Delta \dot{H} \text{ (Open system)} \end{array} \right\} \Delta E_k \text{ and } \Delta E_p \text{ are neglected}$$

- To determine the sensible heat requirement for a heating or cooling process, determine ΔU or $\Delta \dot{H}$ for specified temperature change

Changes in Temperature

- Specific internal energy of a substance depends strongly on variation of specific internal energy with temperature (such that system volume remain constant)



- A temperature change ΔT from T_1 leads to a change $\Delta \hat{U}$ in specific IE

As $\Delta T \rightarrow 0$,

$\left(\frac{\Delta \hat{U}}{\Delta T}\right) \rightarrow$ Approches a limiting value (slope of curve at T_1)

$$C_V(T) = \lim_{\Delta T \rightarrow 0} \left(\frac{\Delta \hat{U}}{\Delta T}\right) = \left(\frac{\partial \hat{U}}{\partial T}\right)_V$$

Changes in Temperature

- Plot of $\hat{U}$ vs T is not a straight line, so C_v (slope of the curve) is a function of temperature
- Change in $\hat{U}$ for a temperature rise from T to $(T + dT)$ at constant volume

$$\Delta\hat{U} = C_v(T) dT$$

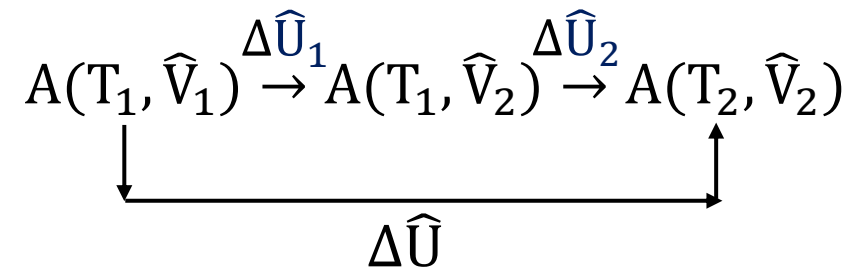
Change $\Delta\hat{U} = \hat{U}_2 - \hat{U}_1$ associated with temperature change from T_1 to T_2 at constant volume is

$$\Delta\hat{U} = \int_{T_1}^{T_2} C_v(T) dT$$

Changes in Temperature

- If both temperature and volume of a substance change, to calculate $\Delta\hat{U}$, break the process into two steps:

A change in V at constant T followed by change in T at constant V



$\hat{U} \rightarrow$ State property

$$\Delta\hat{U} = \Delta\hat{U}_2 + \Delta\hat{U}_1$$

Changes in IE with Temperature

For ideal gases and liquid and solids, $\hat{U}$ depends only on T

In step I : T = Constant $\rightarrow \Delta U_1 \approx 0$ (For ideal gases and liquid and solids)

In step II : T changes at constant volume $\rightarrow \Delta U_2 = \int_{T_1}^{T_2} C_v(T) dT$

For temperature change from T_1 to T_2

$$\Delta \hat{U} \approx \int_{T_1}^{T_2} C_v(T) dT$$

- **Ideal gas: Exact**
- **Solid and liquid: Good approximation**
- **Non-ideal gas: Valid only if V is constant**

Change in Enthalpy at Constant Pressure

Enthalpy depends strongly on temperature

If change in enthalpy resulting from temperature increase from T to $T+\Delta T$ at constant pressure, then

As $\Delta T \rightarrow 0$, $\frac{\Delta \hat{H}}{\Delta T} \rightarrow$ Approaches a limiting value (called heat capacity at constant pressure)

$$C_P(T) = \lim_{\Delta T \rightarrow 0} \left(\frac{\Delta \hat{H}}{\Delta T} \right)$$

$$C_P(T) = \left(\frac{\partial \hat{H}}{\partial T} \right)_P$$

Change in Enthalpy at Constant Pressure

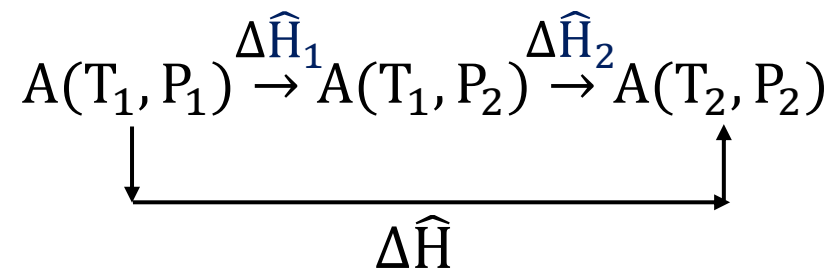
Change in $\hat{H}$ for temperature change at constant pressure from T to $T+\Delta T$ is

$$\Delta\hat{H} = C_p(T) dT$$

For a change from T_1 to T_2 at constant pressure,

$$\Delta\hat{H} = \int_{T_1}^{T_2} C_p(T) dT$$

For a process $A(T_1, P_1) \rightarrow A(T_2, P_2)$: We can constitute a two-step process path



Change in Enthalpy at Constant Pressure

Step I: Change in pressure at constant temperature T

$$\Delta \hat{H} = 0 \text{ (ideal gases)}$$
$$\approx \hat{V} \Delta P \text{ (solids and liquids)}$$

Step II: Change in temperature at constant pressure P, $\Delta \hat{H}_2 = \int_{T_1}^{T_2} C_p(T) dT$

$$\Delta \hat{H} = \Delta \hat{H}_1 + \Delta \hat{H}_2$$

$$\Delta \hat{H} = \int_{T_1}^{T_2} C_p(T) dT$$

➤ **Ideal gas: Exact**

➤ **Nearly Ideal gas: Approximate for variable P**

➤ **Non-ideal gas: Exact only if P constant**

For large pressure change and small T change, $\hat{V} \Delta P$ is dominating relative to the 2nd term

$$\Delta \hat{H} = \hat{V} \Delta P + \int_{T_1}^{T_2} C_p(T) dT$$

➤ **Solid or liquid**

Heat Capacity Formulas

Heat Capacity: C_p and C_v (Physical property of materials, unit: Energy/(Amount-Temp))

Heat capacities are function of temperature:

$$C_p = a + bT + cT^2 + dT^3 \quad (\text{Where, } a, b, c \text{ are given in reference table})$$

Relation between C_p and C_v : $C_p \approx C_v$ (solids and liquids)

$$C_p = C_v + R \quad (\text{Ideal gases})$$

Heat Capacity Formulas

For heating or cooling of any solid, liquid or gas (except a gas far from ideal gas condition), whether or not the pressure changes, $\Delta\hat{H}$ or $\Delta\hat{U}$ can be estimated using expressions:

For gases ($C_P - C_V = R$):

$$\Delta\hat{H} \approx \int_{T_1}^{T_2} C_P(T) dT$$

$$\Delta\hat{U} \approx \int_{T_1}^{T_2} C_V(T) dT \approx \int_{T_1}^{T_2} C_P(T) dT - R(T_2 - T_1)$$

For liquids and Solids ($C_P = C_V$):

$$\Delta\hat{H} \approx \hat{V}\Delta P + \int_{T_1}^{T_2} C_P(T) dT$$

$$\Delta\hat{U} \approx \int_{T_1}^{T_2} C_V(T) dT \approx \int_{T_1}^{T_2} C_P(T) dT$$

Estimation of Heat Capacities

- Polynomial expression for heat capacities (C_p) as a function of temperature are used for the estimation of enthalpy changes.

$$C_p = a + bT + cT^2 + dT^3$$

- Not for all compounds such expressions are available.
- There are rougher approximate methods for estimation of heat capacities in the absence of formula

Estimation of Heat Capacities

Kopp's Rule

- An empirical method **for estimation of heat capacity of a solid or liquid at or near 20°C**
- C_p for molecular compound is the sum of contributions for each element in the compound

Example: Heat capacity of solid Ca(OH)_2

$$(C_p)_{\text{Ca(OH)}_2} = (C_{pa})_{\text{Ca}} + 2 (C_{pa})_{\text{O}} + 2 (C_{pa})_{\text{H}}$$

$$\begin{aligned}(C_p)_{\text{Ca(OH)}_2} &= (26 + 2 \times 17 + 2 \times 9.6) \text{ J/(mol } ^\circ\text{C)} \\ &= 79 \text{ J/(mol } ^\circ\text{C)}\end{aligned}$$

Enthalpy change of a mixture of known composition

$$(C_p)_{\text{mix}}(T) = \sum_{\text{all mixture components}} y_i C_{pi}(T)$$

Where, $(C_p)_{\text{mix}}(T)$ = Heat capacity of the mixture
 y_i = Mass or mole fraction of i^{th} component
 C_{pi} = Heat capacity of i^{th} component

Once $(C_p)_{\text{mix}}$ is known, $\hat{H}$ for change in temperature from T_1 to T_2

$$\Delta \hat{H} = \int_{T_1}^{T_2} (C_p)_{\text{mix}}(T) dT$$

Enthalpy change of a mixture of substances

Rule 1: For a mixture of dissimilar gases or liquids

- Total enthalpy change = Sum of enthalpy changes for the pure mixture components
- For the mixture of gases or dissimilar liquids (for e. g., pentane and hexane), neglecting the effect of enthalpy changes are associated with mixing of components (a good approximation)

Rule 2: For highly dilute solutions of solids or gases in liquids

- Neglect enthalpy change of the solute

Energy Balance in Single-Phase System

Energy balance on processes that do not involve phase changes or chemical reactions:

If the process only involves heating or cooling a single species from T_1 to T_2

1) Evaluate

$$\Delta\hat{U} = \int_{T_1}^{T_2} C_V dT, \quad \Delta\hat{H} = \int_{T_1}^{T_2} C_P dT$$

2) For closed system at constant volume: $\Delta U = n \Delta\hat{U}$

For closed system at constant pressure: $\Delta H = n \Delta\hat{H}$

For open system calculate: $\Delta\dot{H} = \dot{n} \Delta\hat{H}$

Energy Balance in Single-Phase System

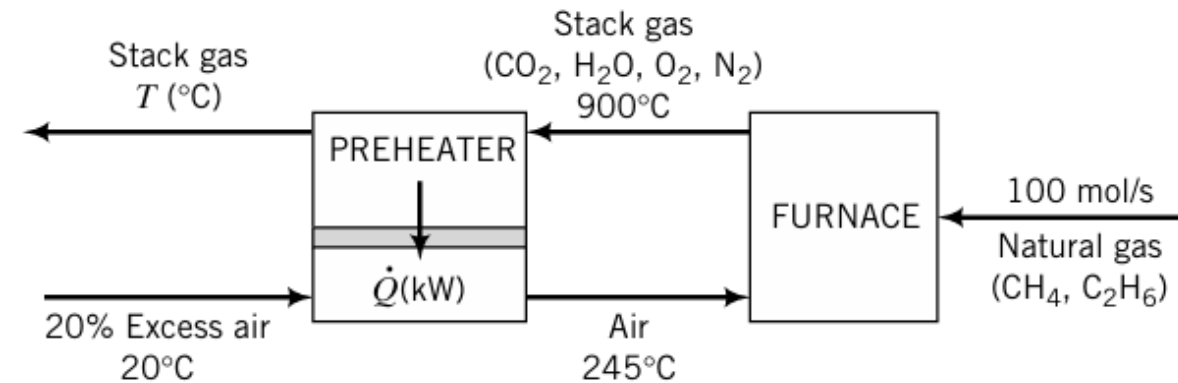
- 3) Substitute ΔU , ΔH , or $\Delta \dot{H}$ in approximate energy balance equation to determine the required heat transfer (Q) or heat transfer rate ($\dot{Q}$)
- 4) If more than one species is involved or if there are several input or output streams instead of just one, procedure is follows:
 - Choose reference state for each species
 - Prepare a table of amounts and specific IE (closed system) or species flow rates and specific enthalpies (open system), and substitute the calculated values into the energy balance

Problem 1

In the manufacture of nitric acid, ammonia and preheated air are mixed to form a gas containing 10.0 mole% NH_3 at 600 deg C. The ammonia is then catalytically oxidized to form NO_2 , which is absorbed in water to form HNO_3 . If ammonia enters the gas blending unit at 25 deg C at a rate of 520 kg/h and heat is lost from the mixer to its surroundings at a rate of 7.00 kW, determine the temperature to which the air must be preheated.

Problem 2

A natural gas containing 95 mole% methane and the balance ethane is burned with 20.0% excess air. The stack gas, which contains no unburned hydrocarbons or carbon monoxide, leaves the furnace at 900 deg C and 1.2 atm and passes through a heat exchanger. The air on its way to the furnace also passes through the heat exchanger, entering it at 20 deg C and leaving it at 245 deg C.



- a) Taking as a basis 100 mol/s of the natural gas fed to the furnace, calculate the required molar flow rate of air, the molar flow rate and composition of the stack gas, the required rate of heat transfer in the preheater, $\dot{Q}$ and the temperature at which the stack gas leaves the preheater.
- (b) What would $\dot{Q}$ be if the actual feed rate of the natural gas were 350 SCMH [standard cubic meters per hour]? Scale up the flowchart of part (a) rather than repeating the entire calculation.

Problem 3

- a) Determine the specific enthalpy (kJ/mol) of n-hexane vapor at 200 deg C and 2.0 atm relative to n-hexane liquid at 20 deg C and 1.0 atm, assuming ideal gas behaviour for the vapor. Show clearly the process path you construct for this calculation and give the enthalpy changes for each step. State where you used the ideal gas assumption.
- b) What is the enthalpy of n-hexane liquid at 20 deg C and 1.0 atm relative to n-hexane vapor at 200 deg C and 2.0 atm?
- c) Beginning with the value of $\hat{H}$ calculated in part (a) and still assuming ideal gas behaviour, determine the specific internal energy of the vapor at 200 deg C and 2.0 atm. Once again, state where you used the ideal gas assumption.

Problem 4

Benzene vapor at 580 deg C is cooled and converted to a liquid at 25 deg C in a continuous condenser. The condensate is drained into 1.75-m³ drums, each of which takes 2.0 minutes to fill. Calculate the rate (kW) at which heat is transferred from the benzene in the condenser.

Problem 5

If carbon dioxide is cooled at 1 atm, it condenses directly to a solid (dry ice) at -78.4 deg C . The heat of sublimation at this temperature is $\Delta\hat{H}_{sub}(-78.4\text{ deg C}) = 6030\text{ cal/mol}$.

Calculate the heat removal rate (kW) required to produce 300 kg/h of dry ice at 1 atm and -78.4 deg C if $\text{CO}_2(\text{v})$ at 20 deg C is the feed.

Problem 6

Seventy-five liters of liquid ethanol at 70.0 deg C and 55 L of liquid water at 20.0 deg C are to be mixed in a well-insulated flask. The energy balance for this constant pressure process is $Q = \Delta H$.

- a) As part of the calculation, use data in Table B.2 to estimate a linear formula for the heat capacity of liquid ethanol.
- b) Neglecting evaporation and the heat of mixing, estimate the final mixture temperature.

Problem 7

A gas stream containing 8.0 mole% CO and 92.0 mole% CO₂ at 500 deg C is fed to a large metal shell containing a bundle of small-diameter tubes. The hot gas flows over the outside of the tubes. Liquid water at 25 deg C is fed to the boiler in a ratio 0.200 mol feedwater/mol hot gas and flows inside the tubes. Heat is transferred from the hot gas through the tube walls to the water, causing the gas to cool and the water to heat to its boiling point and evaporate to form saturated steam at 5.0 bar. The steam may be used for heating or power generation in the plant or as the feed to another process unit. The gas leaving the boiler is flared (burned) and discharged to the atmosphere.

The boiler operates adiabatically—all the heat transferred from the gas goes into the water, as opposed to some of it leaking through the outside boiler wall.

Assume the basis of 1.00 mol feed gas. What is the temperature of the exiting gas?

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Phase Change Operation

Consider liquid water and water vapor, each at 100°C and 1 atm. Which would you expect to be greater, $\hat{U}_{\text{liq}}$ or $\hat{U}_{\text{vap}}$?

Water at 100°C and 1 atm: Specific Internal Energy

$$\hat{U}_l = 419 \text{ kJ/mol}$$

$$\hat{U}_v = 2507 \text{ kJ/mol}$$

Water at 100°C and 1 atm: Specific Enthalpy

$$\hat{H}_l = 419.1 \text{ kJ/mol}$$

$$\hat{H}_v = 2656 \text{ kJ/mol}$$

Phase Change Operation

Different types of phase change operation

- Phase changes in liquid and vapor: **Evaporation and Condensation**
- Phase changes in solid and liquid: **Melting and Freezing**
- Phase change from solid to vapor: **Sublimation**
- Phase change from one solid phase to another

Latent Heats

- **Latent heat-** Associated with specific enthalpy change, associated with the transition of a substance from one phase to another at constant T and P

Ex: Sp. enthalpy change for transition of liquid water to steam @ 100°C and 1 atm

$$-\Delta\hat{H}_V = 40.6 \text{ kJ/mol}$$

Heat of condensation at 100°C and 1 atm = - 40.6 kJ/mol

$$= -\Delta\hat{H}_V$$

- Condensation is reverse of vaporization and enthalpy is state property

Latent Heats

- Latent heat for phase changes:

Heat of fusion (or, heat of melting) = $\Delta\hat{H}_m(T, P)$

Heat of solidification = $-\Delta\hat{H}_m(T, P)$

- Latent heat of phase change vary considerably with temperature at which the change occurs, but hardly varies at all with the pressure at the transition point

For e.g., $-\Delta\hat{H}_V(\text{H}_2\text{O}, 25^\circ, 23.78 \text{ mm Hg}) = 2442.5 \text{ J/g}$

$-\Delta\hat{H}_V(\text{H}_2\text{O}, 25^\circ, 760 \text{ mm Hg}) = 2442.3 \text{ J/g}$

Estimation and Correlation of Latent Heats

Trouton's rule: (Accurate to within 30 %)

Estimate standard heat of vaporization $\Delta\hat{H}_V$ at normal boiling point

$$\begin{aligned}\Delta\hat{H}_V &\approx 0.088 T_b(\text{K}) \\ &\approx 0.109 T_b(\text{K})\end{aligned}$$

Chen's equation: (Accurate to within 2 %)

$$\Delta\hat{H}_V \approx \frac{T_b[0.0331(T_b/T_c) - 0.0327 + 0.0297 \log_{10} P_c]}{1.07 - (T_b/T_c)}$$

Standard heat of fusion

$$\begin{aligned}\Delta\hat{H}_m &= 0.0092 T_m \quad (\text{metallic elements}) \\ &= 0.0025 T_m \quad (\text{Inorganic compounds}) \\ &= 0.05 T_m \quad (\text{Organic compounds})\end{aligned}$$

Estimation and Correlation of Latent Heats

Clapeyron Equation

Latent heat of vaporization estimated from vapor pressure data if it varies with temperature

$$\frac{d(\ln(P^*))}{d(1/T)} = \frac{-\hat{H}_V}{RT}, \quad \Delta\hat{H}_V \text{ is determined from the plot } \ln(P^*) \text{ vs } (1/T)$$

Clausius-Clapeyron Equation

If latent heat of vaporization of a substance is independent of temperature over which vapor pressure is available

$$\ln(P^*) = \frac{-\hat{H}_V}{RT} + B$$

Plot $\ln(P^*)$ vs $(1/T)$ -> determine slope $d(\ln(P^*)) / d(1/T)$ at temperature of interest of the tangent to the curve

Estimation and Correlation of Latent Heats

Watson's Correlation

An approximation to estimate $\Delta\hat{H}_V$ at T_2 from known value of T_1

$$\Delta\hat{H}_V(T_2) = \Delta\hat{H}_V(T_1) \left(\frac{T_C - T_2}{T_C - T_1} \right)^{0.38}$$

Energy Balance for Phase Change

- **Energy balance for a process in which a component exists in two phases**
 - Choose a reference state for that component by specifying both a phase and a temperature and calculate specific enthalpy of the component in all process streams relative to this state
- **If substance is a liquid at its reference state and vapor in a process stream, then for calculating $\hat{H}$**
 - Bring liquid from reference temperature to a point at which $\Delta\hat{H}_V$ is known
 - Vaporize the liquid
 - Bring the vapor to process stream temperature
 - Sum the individual enthalpy changes for the three steps

Problem 1

Molten sodium chloride is to be used as a constant-temperature bath for a high-temperature chemical reactor. Two hundred kilograms of solid NaCl at 300K is charged into an insulated vessel, and a 3000 kW electrical heater is turned on, raising the salt to its melting point of 1073 K and melting it at a constant pressure of 1 atm.

- The heat capacity (C_p) of solid NaCl is 50.41 J/(mol.K) at 300 K, and 53.94 J/(mol.K) at 500 K, and the heat of fusion of NaCl at 1073 K is 30.21 kJ/mol. Use these data to determine a linear expression for $C_p(T)$ and to calculate $\Delta\hat{H}$ (kJ/mol) for the transition of NaCl from a solid at 300 K to a liquid at 1073 K.
- Write and solve the energy balance equation for this closed system isobaric process to determine the required heat input in kilojoules.
- If 85% of the full power of 3000 kW goes into heating and melting the salt, how long does the process take?

Problem 2

You are writing energy balances for a compound for which you can not find heat capacity or latent heat data. All you know about the material are its molecular formula ($C_7H_{12}N$) and that it is a liquid at room temperature and has a normal boiling point of 200 deg C. Use this information to estimate the enthalpy of the vapor of this substance at 200 deg C relative to the liquid at 25 deg C

Problem 3

A stream of pure cyclopentane vapor flowing at a rate of 1550 L/s at 150 deg C and 1 atm enters a cooler in which 55% of the feed is condensed at constant pressure.

- a) What is the temperature at the condenser outlet?
- b) Prepare and fill in an inlet–outlet enthalpy table and calculate the required cooling rate in kW.

Problem 4

Estimate the heat of vaporization (kJ/mol) of benzene at a pressure of 100 mm Hg, using each of the following correlations and data:

- a) The heat of vaporization at the normal boiling point given in Table B.1, the boiling point at 100 mm Hg as determined from the Antoine equation, and Watson's correlation.
- b) The Clausius–Clapeyron equation and the boiling points at 50 mm Hg and 150 mm Hg as determined from the Antoine equation.
- c) The heat of vaporization at the normal boiling point given in Table B.1, the boiling point at 100 mm Hg as determined from the Antoine equation, and heat capacity data given in Table B.2.

BBL 133 Mass and Energy Balance

CHAPTER 2

Energy Balances on Nonreactive Processes

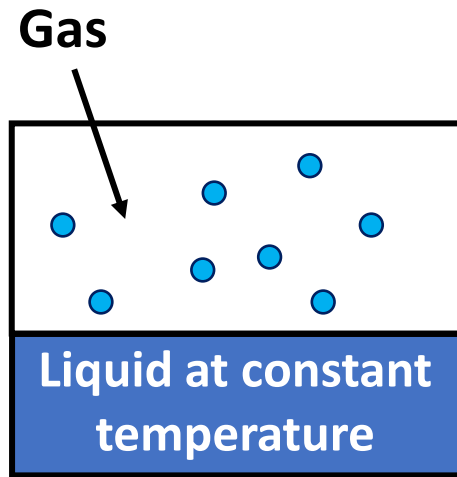
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Psychrometric Chart or Humidity Chart

- Most common humidity chart (air-water system at 1 atm): Used extensively in the analysis of humidification, drying, and air conditioning process
- Provides different physical properties (such as absolute humidity, relative humidity, dry and wet bulb temperature, dew point, humid volume) of humid air at 1 atm appear on humidity chart
- If the value of any of two of these properties are known, humidity chart can be used to determine the value of others

Humid Air and Related Terminologies



Gas acquire vapor from the liquid until the partial pressure of vapor in the gas mixture equals the vapor pressure of the liquid at its existing temperature

Partial saturation

$$\text{Relative saturation} = \frac{\text{Partial pressure of vapor} \times 100}{\text{Vapor pressure of pure liquid}}$$

Percentage saturation: measure of degree of saturation of vapor-bearing gas

$$\% \text{ saturation} = \frac{\text{moles of vapor per moles of vapor free gas actually present}}{\text{moles of vapor per mole of vapor free gas if the mixture were saturated at the existing T and P}} \times 100$$

Relative and % saturation approach equality when the vapor concentration approaches zero, but are different at all other conditions

Humid Air and Related Terminologies

Humidity of gas or air

- Weight of water per unit weight of moisture free gas
- Concentration of water vapor (gaseous state of water) present in the air
- MOLAR HUMIDITY: Number of moles of water per mole of moisture free gas

Absolute Humidity (Moisture Content)

- Abs humidity of a gas is defined as weight of water vapor per unit weight of moisture free gas

$$h_a = \frac{\text{kg H}_2\text{O (v)}}{\text{kg Dry Air}}$$

Humid Air and Related Terminologies

Relative humidity and Percentage Humidity

- When the vapor under consideration is water, the % saturation is called % humidity and relative saturation is called relative humidity

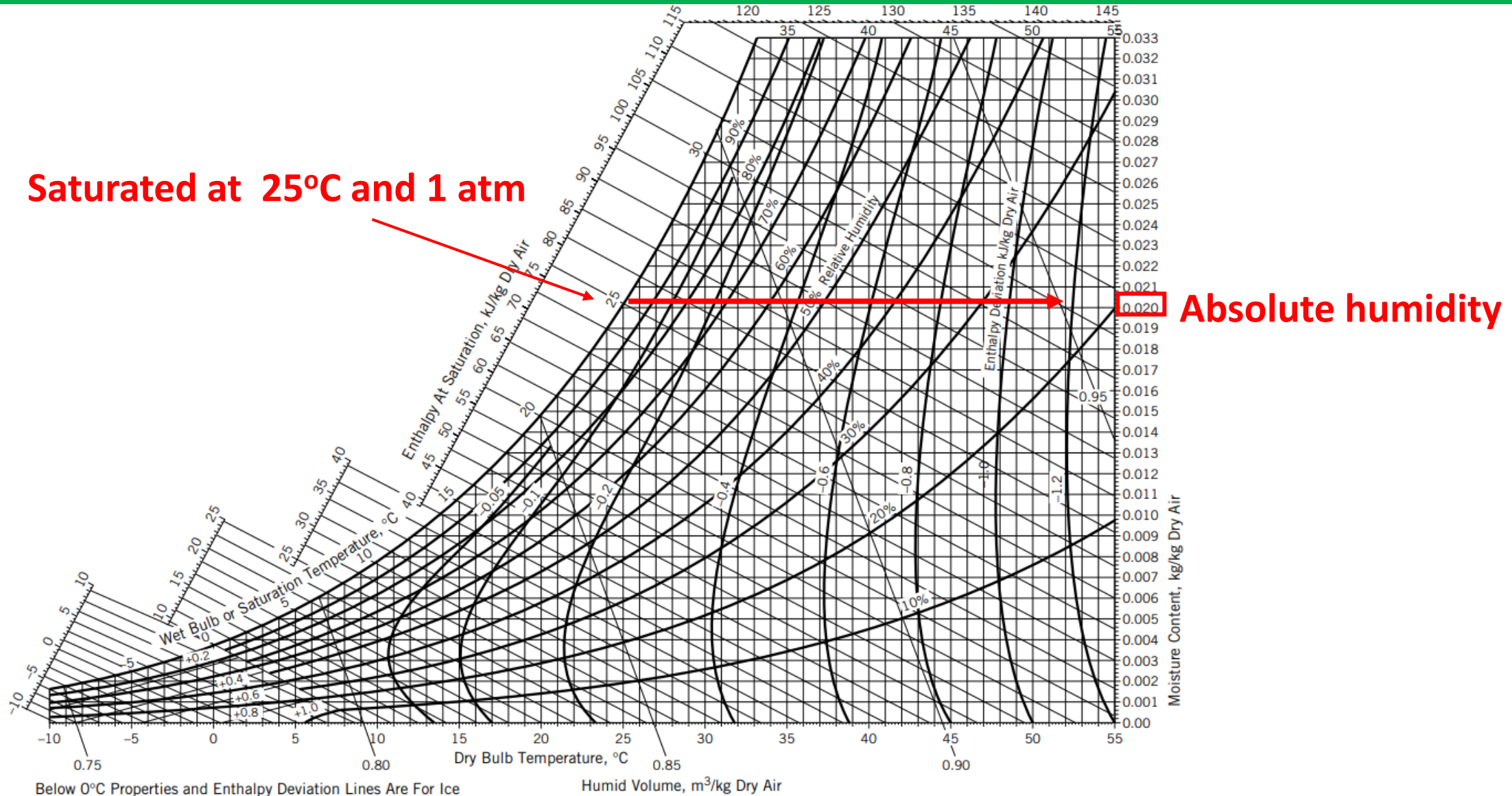
$$h_r = \frac{\text{Partial pressure of water vapor}}{\text{Equilibrium vapor pressure of water}} \times 100 = \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2\text{O}}^*(T)} \times 100$$

Example: If $h_a = 0.015 \text{ kg H}_2\text{O/kg DA}$ $\Rightarrow$ For every kg of DA, there is 0.015 kg of water vapor

For a total of 1.015 kg Humid Air

$$\text{Mass fraction of water} = \frac{0.015 \text{ kg H}_2\text{O}}{1.015 \text{ kg Humid Air}} = 0.0148 \frac{\text{kg H}_2\text{O}}{\text{kg Humid Air}}$$

Relative and Absolute Humidity



Humid Air and Related Terminologies

Humid Volume $\left(\hat{V}_H \left[\frac{\text{m}^3}{\text{kg Dry Air}} \right] \right)$

- Volume occupied by 1 kg of dry air plus the water vapor that accompanies it
- The volume of a given mass of wet air can be determined by determining corresponding mass of dry air from abs humidity, then multiply this mass by $\hat{V}_H$

Example:

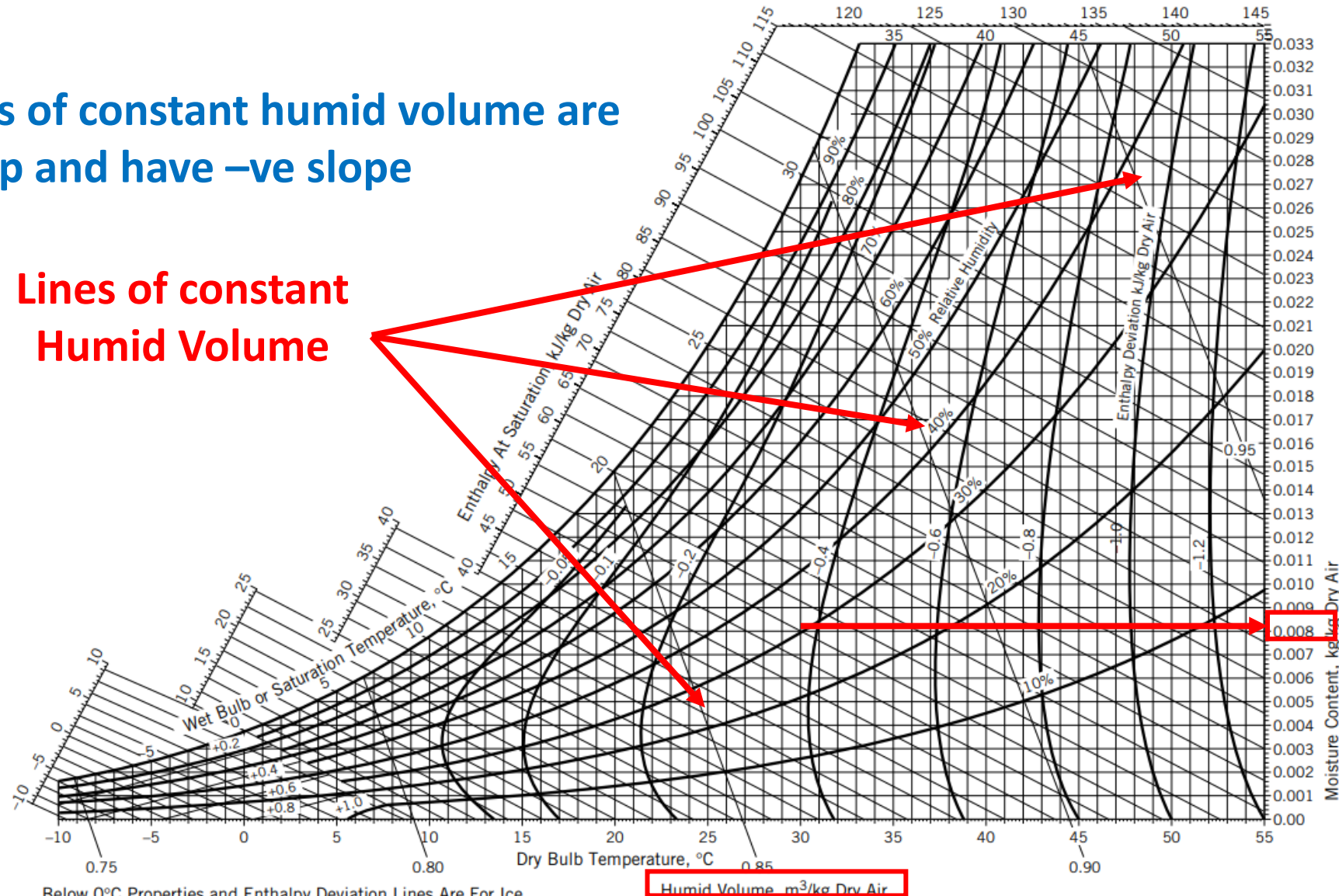
150 kg humid air at $T = 30^\circ\text{C}$, $h_r = 30\%$ ($h_a = 0.008 \text{ kg H}_2\text{O(v)}/\text{kg DA}$, $\hat{V}_H = 0.87 \text{ m}^3/\text{kg DA}$)

$$\text{Volume, } V = 150 \text{ kg Humid Air} \times \frac{1 \text{ kg DA}}{1.008 \text{ kg Humid Air}} \times \frac{0.87 \text{ m}^3}{\text{kg DA}} = 129 \text{ m}^3$$

Humid Volume

- Lines of constant humid volume are steep and have –ve slope

Lines of constant
Humid Volume



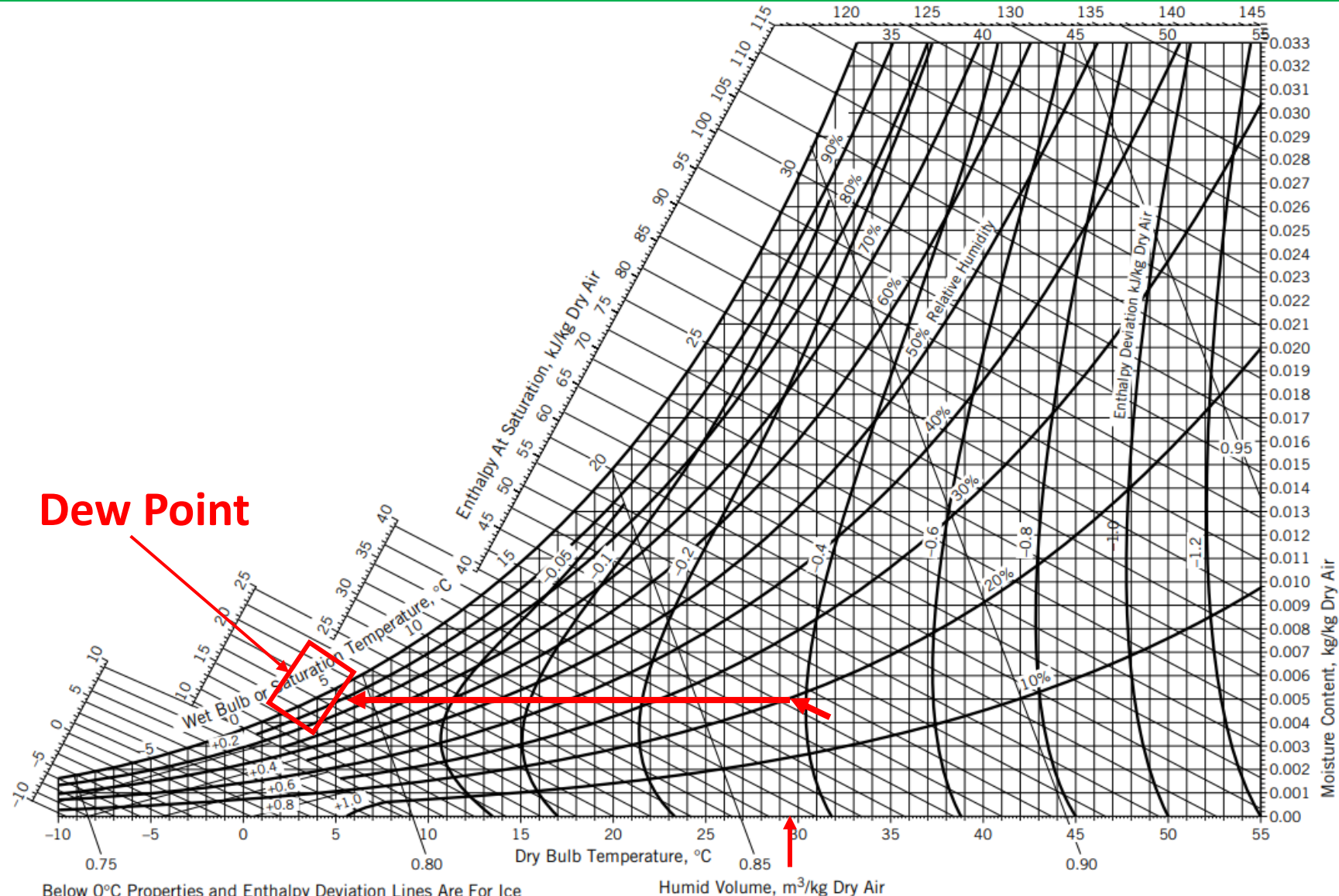
Humid Air and Related Terminologies

Dew Point (T_{dp})

- Temperature at which humid air becomes saturated with water vapor if it is cooled at constant pressure
- In the humidity chart, locate the point on the chart corresponding to air temperature and % relative humidity. Cooling of this air at constant P corresponds to moving horizontally (at constant absolute humidity) to the saturation curve. T_{dp} is the temperature at this intersection.

Example: Air at 29°C and 20% relative humidity. Cooling this air at constant pressure (1 atm) corresponds to $T_{dp} = 4^\circ\text{C}$

Dew Point



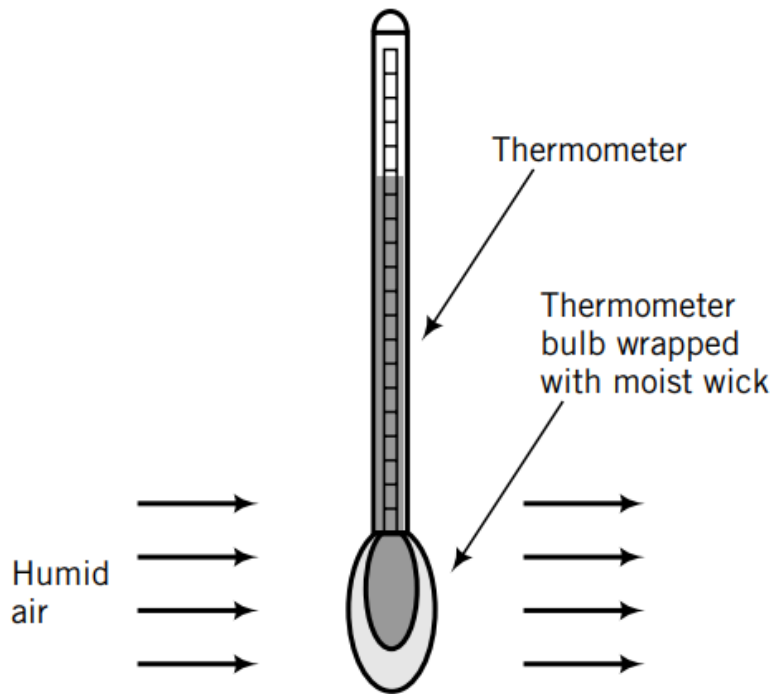
Humid Air and Related Terminologies

Wet Bulb Temperature (T_{wb}) of Humid Air

- If the body of liquid has a large surface area in proportion to its mass, its temperature will quickly drop to an equilibrium value
- The equilibrium temperature attained by liquid or air when a balance established between the **rate of heat transfer to the liquid from the warmer ambient gas and rate of transfer of vapor from the liquid to the ambient gas**, is Wet Bulb Temperature
- The wet bulb temperature is always less than the actual dry-bulb temperature of the gas into which evaporation is taking place

Wet Bulb Temperature

Wet Bulb Temperature

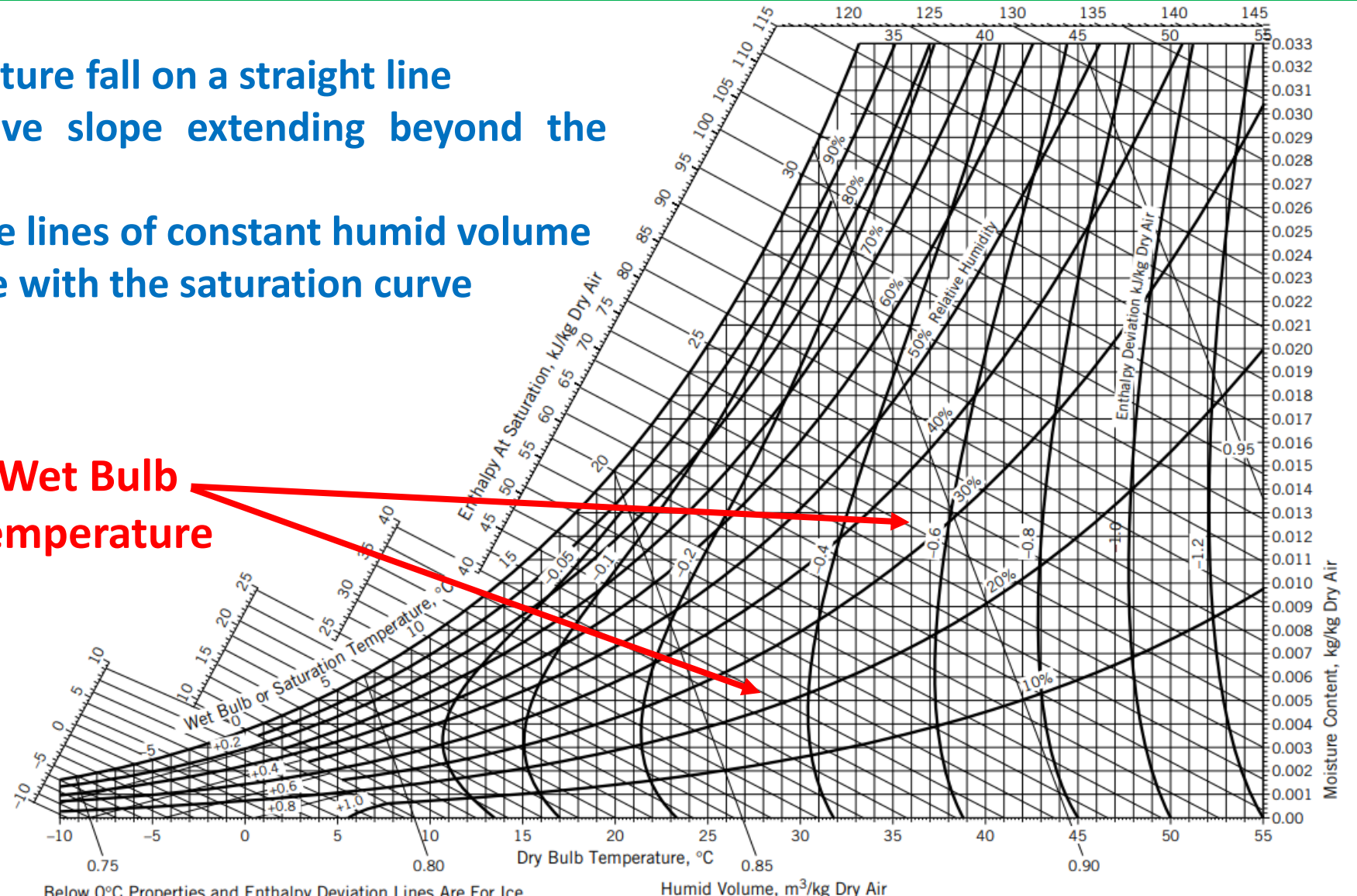


- The wet-bulb temperature depends on both, the dry-bulb temp and the moisture content of the air
- For saturated air (100% relative humidity), no water evaporates from the wick, and the wet bulb and dry bulb temperatures are same
- The lower the humidity, the greater the difference between the two temperatures.

Wet Bulb Temperature

- Wet-bulb temperature fall on a straight line
- Lines with negative slope extending beyond the saturation curve
- Less steep than the lines of constant humid volume
- Intersection of line with the saturation curve

**Wet Bulb
Temperature**



Specific Enthalpy of Saturated Air

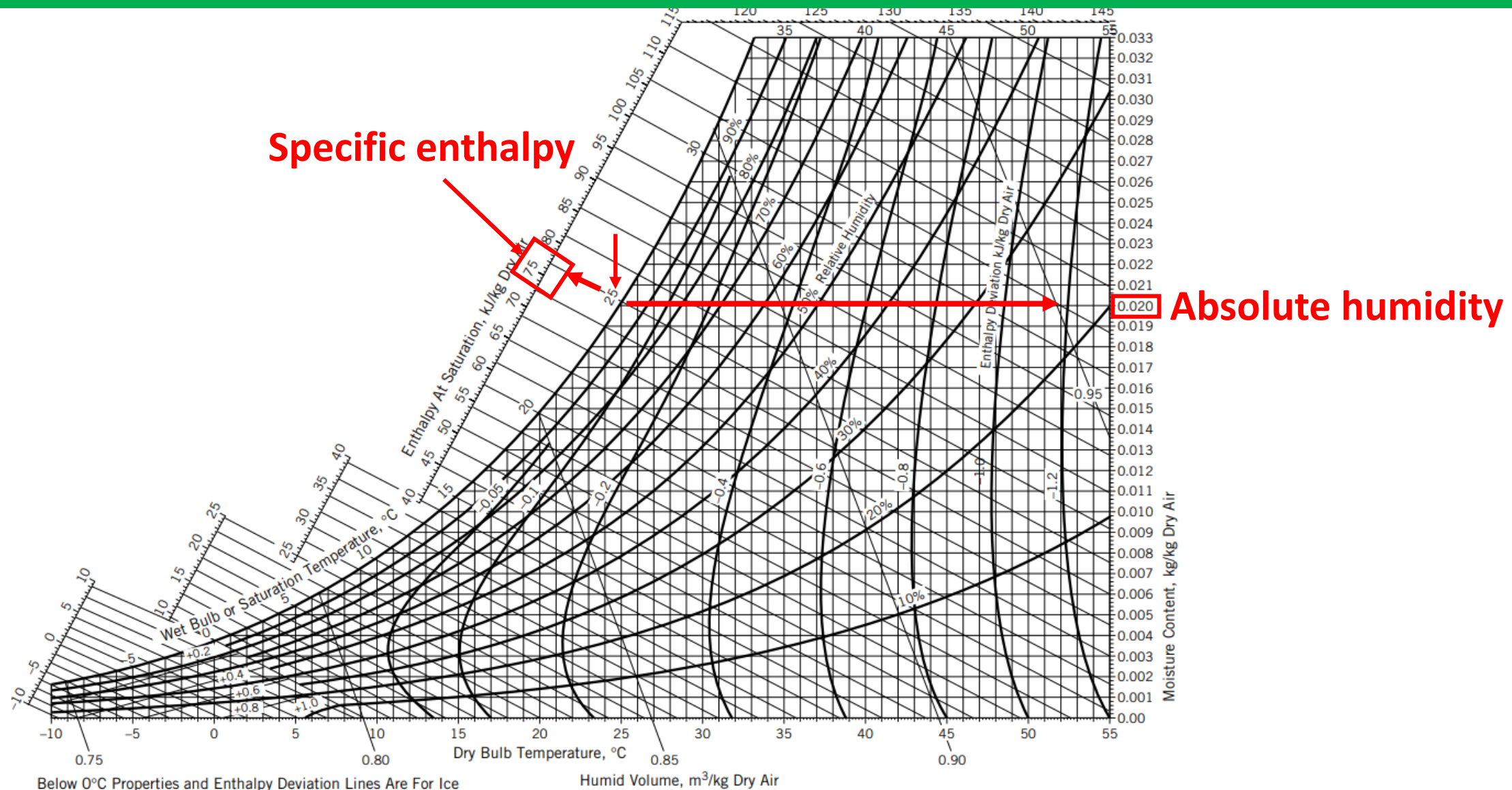
- Enthalpy of a unit mass (1 kg or 1 lb_m) of dry air plus water vapor is present above the saturation curve (the diagonal scale)
- References states: Liquid water at 1 atm and 0°C (32°F), or DA at 1 atm and 0°C (32°F)
- To determine enthalpy from the chart, follow the constant wet-bulb temperature line from the saturation curve at the desired temperature up to the enthalpy scale

Example: Given saturated air at 25°C and 1 atm,

Absolute humidity $h_a = 0.0202$ kg H₂O/kg DA

Specific enthalpy $\hat{H} = 76.5$ KJ/kg DA

Specific Enthalpy on Humidity Chart



Specific Enthalpy of Saturated Air

Total Enthalpy = (Enthalpy changes for 1 kg dry air + 0.0202 kg water) going from their reference condition to 25°C

1 kg DA (0°C, 1 atm) → 1 kg DA (25°C, 1 atm)

$$\Delta H_{\text{air}} = (1 \text{ kg DA}) \left(\frac{1 \text{ kmol}}{25 \text{ kg}} \right) \int_0^{26} C_{p,\text{air}}(T) dt = 25.1 \text{ kJ}$$

0.0202 kg H₂O (l, 0°C) → 0.0202 kg H₂O (v, 25°C)

$$\Delta H_{\text{water}} = (0.0202 \text{ kg}) \left[\hat{H}_{\text{H}_2\text{O}}(\text{v}, 25^\circ\text{C}) - \hat{H}_{\text{H}_2\text{O}}(\text{l}, 0^\circ\text{C}) \right] = 51.4 \text{ kJ}$$

$$\hat{H} = \frac{(\Delta H_{\text{air}} + \Delta H_{\text{water}}) \text{ kJ}}{1 \text{ kg DA}} = \frac{(25.1 + 51.4) \text{ kJ}}{1 \text{ kg DA}} = 76.5 \frac{\text{kJ}}{\text{kg DA}}$$

Problem 1

On a cold winter day the temperature is 2 deg C and the relative humidity is 15%. You inhale air at an average rate of 5500 mL/min and exhale a gas saturated with water at body temperature, roughly 37 deg C. If the mass flowrates of the inhaled and exhaled air (excluding water) are the same, the heat capacities (C_p) of the water-free gases are each 1.05 J/(g. deg C), and water is ingested into the body as a liquid at 22 deg C, at what rate in J/day do you lose energy by breathing? Treat breathing as a continuous process (inhaled air and liquid water enter, exhaled breath exits) and neglect work done by the lungs.

Problem 2

In gas adsorption a vapor is transferred from a gas mixture to the surface of a solid. An approximate but useful way of analyzing adsorption is to treat it simply as condensation of vapor on a solid surface.

Suppose a nitrogen stream at 35 deg C and 1 atm containing carbon tetrachloride with a 15% relative saturation is fed at a rate of 10.0 mol/min to a 6-kg bed of activated carbon. The temperature and pressure of the gas do not change appreciably from the inlet to the outlet of the bed, and there is no CCl₄ in the gas leaving the adsorber. The carbon can adsorb 40% of its own mass of carbon tetrachloride before becoming saturated.

Neglecting the effect of temperature on the heat of vaporization of CCl₄, estimate the rate at which heat must be removed from the adsorber (kJ/min) to keep the process isothermal and the time (min) it will take to saturate the bed.

Problem 3

An air conditioner cools 226 m³/min of humid air at 36 deg C and 98% relative humidity to 10 deg C.

(a) Do a degree-of-freedom analysis to prove that enough information is available to determine the required cooling duty (rate of heat transfer).

(b) Calculate the rate of condensation of water in the unit and the cooling duty in tons (1 ton 12,000 Btu/h).

BBL 133 Mass and Energy Balance

CHAPTER 2

Energy Balances on Nonreactive Processes

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Mixing and Solution

For mixing of pure H_2SO_4 with water at a specified temperature and pressure, and then cool the mixture at constant pressure to bring it back to the initial temperature

The energy balance for these constant pressure process

$$Q = \Delta H = H_{\text{H}_2\text{SO}_4(\text{aq})} - (H_{\text{H}_2\text{SO}_4(\text{l})} + H_{\text{H}_2\text{O}}) = \textbf{Heat of solution}$$

For H_2SO_4 solution, $Q < 0$ (the container must be cooled to keep the solution temperature from rising) and it follows that the heat of solution (ΔH) is –ve for this process.

Heat of Solution and Mixing

Heat of solution ($\Delta\hat{H}_s(T, r)$)

- Change in enthalpy for a process in which **1 mole of a solute (gas or solid) is dissolved in r moles of a liquid solvent** at a constant temperature T
- As r becomes large, $\Delta\hat{H}_s$ approaches a limiting value K/A heat of solution at infinite dilution
- $\Delta\hat{H}_s$ of various substance at specified temperature can be find in Perry's handbook

Heat of mixing

- When the **process involves mixing two fluids rather than dissolving a gas or solid in a liquid**

Heat of Mixing

Example of Heat of Mixing:

2 mol of KCN is dissolved in 400 mol of water at 18°C. Calculate ΔH of this process.

Moles of solvent per mole of solute, $r = 400/2 = 200$ mol H₂O/mol KCN

From reference data, @18°C and $r = 200$: - $\Delta H_s = - 3$ kcal/mol

Total enthalpy change,

$$\Delta H = n \Delta \hat{H}_s = (2 \text{ mol KCN}) \times 3 \frac{\text{kcal}}{\text{mol KCN}}$$

$$\Delta H = 6 \text{ kcal}$$

Heat of Solution and Mixing

How to choose reference condition?

- 1) If heat of solution of **pure substances are given in reference data at 25°C**, can be used to determine specific enthalpy of solution relative to pure solute and solvent at this temperature
- 2) Reference condition is **pure solvent and an infinitely dilute solution at 25°C**

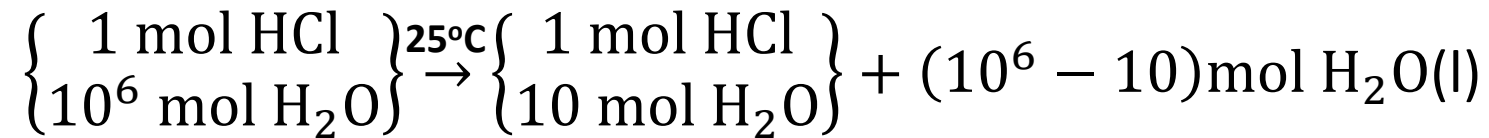
Example: Given a HCl solution for which $r = 10$ moles $\text{H}_2\text{O}/\text{mol HCl}$. Calculate enthalpy change.

From reference data $\hat{H}$ of this solution relative to pure solute HCl (g) and solvent H_2O (l) at 25°C is

$$\Delta\hat{H}_s(r = 10) = -69.49 \frac{\text{kJ}}{\text{mol}} \text{HCl}$$

Heat of Solution and Mixing

- 1) Enthalpy change can be evaluated, either using reference state – **pure HCl (g) and H₂O (l) @25°C**, or
- 2) Enthalpy of solution relative to **pure solvent H₂O (l) and a highly dilute solution of HCl** (e.g., $r = 10^6$ moles H₂O/mol HCl) is the enthalpy change for the isothermal process



For choosing pure water as reference, $\hat{H}$ for $(10^6 - 10)$ moles of pure water equals zero, and the enthalpy change for the process:

$$\Delta\hat{H} = \Delta\hat{H}_s(r = 10) - \Delta\hat{H}_s(r = \infty) = \frac{(-69.49 + 75.14)\text{kJ}}{\text{mol}} \text{HCl} = 5.65 \frac{\text{kJ}}{\text{mol}} \text{HCl}$$

Heat of Solution and Mixing

- In general, **for reference state of pure solute and solvent at 25°C and 1 atm**, enthalpy change of a solution containing r moles $\text{H}_2\text{O}/\text{mol}$ solute is

$$\hat{H} = \Delta\hat{H}_s(r)$$

- In general, **for reference state of pure solvent and an infinitely dilute solution at 25°C and 1 atm**, the enthalpy change of a solution containing r moles $\text{H}_2\text{O}/\text{mol}$ solute

$$\hat{H} = \Delta\hat{H}_s(r) - \Delta\hat{H}_s(\infty)$$

Energy Balances on dissolution & mixing Processes

- In energy balance problems, where heat of solution or mixing can't be neglected, an inlet-outlet enthalpy table is required by considering the solution as a single substance and the pure component at 25°C as reference states.
- To calculate enthalpy of solution at $T \neq 25^\circ\text{C}$, first calculate its enthalpy at 25°C from tabulated heat of solution data, and then add the enthalpy change for the heating or cooling of the solution from 25°C to T
- For calculating enthalpy change, solution heat capacities can be obtained
 - Reference data if available
 - Average heat capacity: $(C_p)_{\text{mix}}(T) = \sum_{\text{all mixture component}} y_i C_{pi}(T)$ for liquid mixture
 - Heat capacity of the pure solvent for dilute solution

Problem 1

Use Table B.11 to determine the specific enthalpy (kJ/mol HCl) of hydrochloric acid containing 1 mol HCl/5 mol H₂O at 25 deg C relative to:

- A) HCl(g) and H₂O(l) at 25 deg C.
- B) H₂O(l) and an infinitely dilute HCl solution at 25 deg C.

Problem 2

Sodium hydroxide is dissolved in enough water to make up a 20.0 mole% solution. If the NaOH and water are initially at 77 deg F (25 deg C), how much heat (Btu/lb product solution) must be removed for the solution also to be at 77 deg F. Assume the process is carried out at constant pressure, so that $Q = \Delta H$, and use Table B.11 to evaluate .

Problem 3

A sulfuric acid solution is labelled 8N (where 1N 1g-equivalent/L, and 1 mol of H_2SO_4 contains two g-equivalents). The specific gravity of the solution is 1.230, and its heat capacity is $3.00 \text{ J}/(\text{g} \cdot \text{deg C})$. Calculate the specific enthalpy of this solution (in $\text{kJ}/\text{mol H}_2\text{SO}_4$) at 60 deg C relative to pure H_2O and an infinitely dilute solution at 25 deg C .

BBL 133 Mass and Energy Balance

CHAPTER 2

Energy Balances on Nonreactive Processes

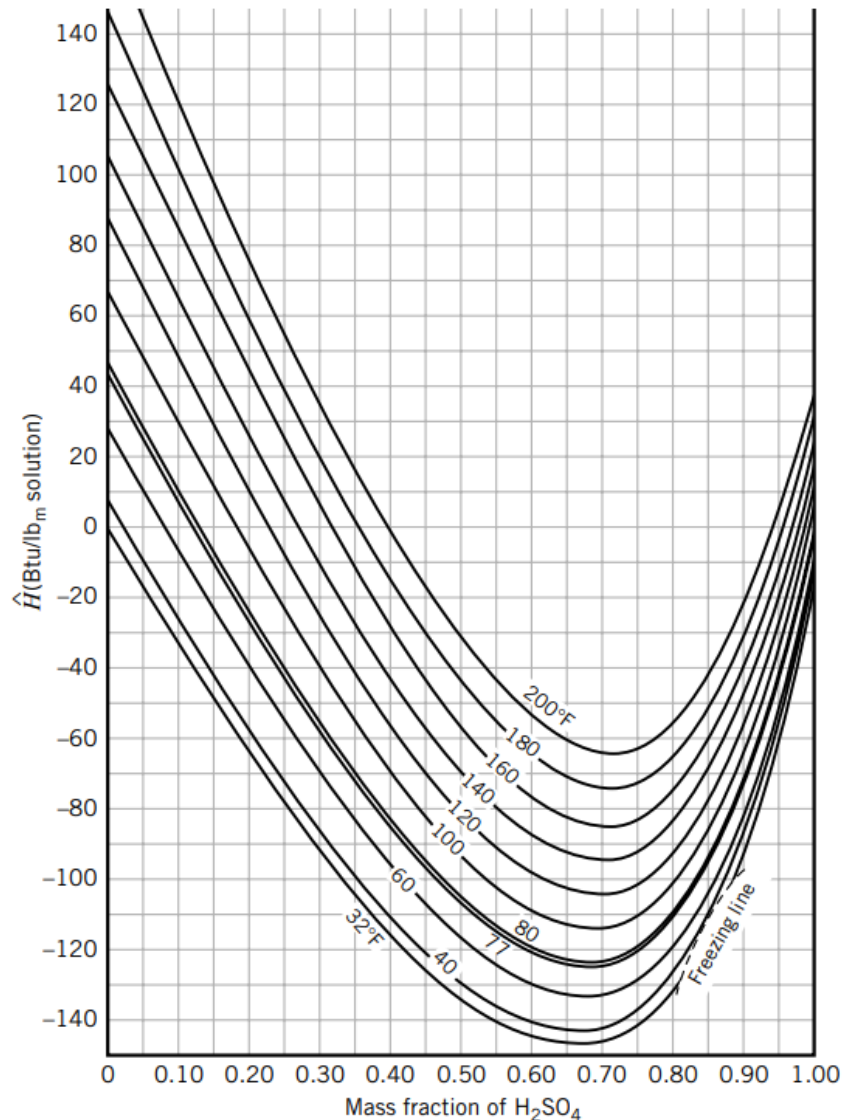
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Enthalpy-Conc Charts: Single Phase Liquid

- When heat of mixing are significant, energy balance calculations on liquid-phase systems involving mixtures can be complicated
- The calculation can be simplified for binary (two component system) by using an enthalpy-concentration chart, a plot of sp. enthalpy vs mole fraction (or mole%) or mass fraction (or wt%) of one component

Enthalpy-Conc Charts for H_2SO_4 - H_2O



- H-x chart of aqueous solution of H_2SO_4 at several temperature
- Reference conditions:
 - Pure liquid H_2SO_4 at 77°F
 - Liquid water at 32°F

Enthalpy-Conc Charts: Single Phase Liquid

○ **Calculate specific enthalpy (Btu/lb_m) of a 40 wt% H₂SO₄ solution at 120°F**

- Heat of mixing of H₂SO₄ at 77°F

- Process path:

- Bring pure liquid water from its reference temperature of 32°F to 77°F
- Mix the two liquids at 77°F
- Bring the product solution to 120°F
- Calculate and add the enthalpy change for each of these steps

$$\begin{array}{lcl} \text{Basis: 1 lb}_m \text{ solution} & \begin{array}{l} \nearrow \\ \searrow \end{array} & \begin{array}{l} 0.4 \text{ lb}_m \text{ H}_2\text{SO}_4 = 4.08 \times 10^{-3} \text{ lb-mole} \\ 0.6 \text{ lb}_m \text{ H}_2\text{O} = 3.33 \times 10^{-2} \text{ lb-mole} \end{array} \end{array}$$

Enthalpy-Conc Charts: Single Phase Liquid

➤ $0.6 \text{ lb}_m \text{ H}_2\text{O (l, 32°F)} \rightarrow 0.6 \text{ lb}_m \text{ H}_2\text{O (l, 77°F)}$

$$\Delta H_1 (\text{Btu}) = (0.6 \text{ lb}_m \text{ H}_2\text{O}) \int_{32^\circ\text{F}}^{77^\circ\text{F}} (C_P)_{\text{H}_2\text{O}} dT$$

➤ $0.4 \text{ lb}_m \text{ H}_2\text{SO}_4 (77^\circ\text{F}) + 0.6 \text{ lb}_m \text{ H}_2\text{O (77°F)} \rightarrow 1 \text{ lb}_m \text{ H}_2\text{SO}_4 (\text{aq, 77°F})$

$$\Delta H_2 (\text{Btu}) = (0.4 \text{ lb}_m \text{ H}_2\text{SO}_4) \left[\Delta \hat{H}_s \left(77^\circ\text{F}, r = 8.2 \frac{\text{lb mole H}_2\text{O}}{\text{lb mole H}_2\text{SO}_4} \right) \right]$$

Heat of mixing can be determined from table = $-279 \text{ Btu/lb}_m \text{ H}_2\text{SO}_4$

Enthalpy-Conc. Charts-Single Phase Liquid

➤ 1 lb_m H₂SO₄ (aq., 77°F) → 1 lb_m H₂SO₄ (aq., 120°F)

$$\Delta H_3(\text{Btu}) = (1.0 \text{ lb}_m) \int_{77^\circ\text{F}}^{120^\circ\text{F}} (C_P)_{40\% \text{H}_2\text{SO}_4} dT$$

Heat capacity of 40% H₂SO₄ solution = 0.67 Btu/(lb_m °F)

➤ $\hat{H}(40\% \text{ H}_2\text{SO}_4, 120^\circ\text{F}) = \frac{(\Delta H_1 + \Delta H_2 + \Delta H_3)}{1 \text{ lb}_m \text{ solution}} = -56 \text{ Btu/lb}_m$

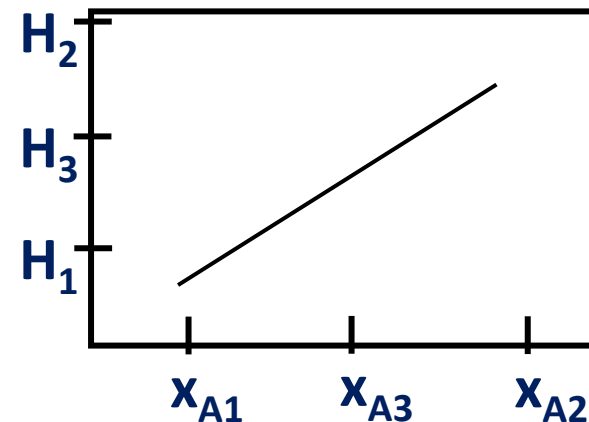
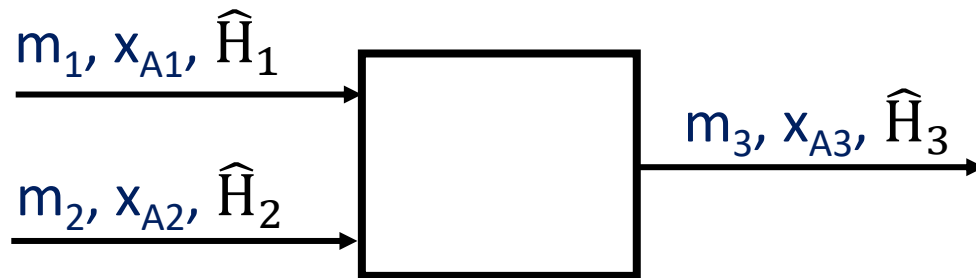
Adiabatic mixing

Consider a mixture of two species, A1 and A2, formed by mixing two solutions adiabatically.

m_1 : Mass of solution 1 ($x_{A1}, \hat{H}_1$)

m_2 : Mass of solution 2 ($x_{A2}, \hat{H}_2$)

Find condition for the produce mixture ($x_{A3}, \hat{H}_3$) to be on a straight line on the H-X chart between the points corresponding to the feed stream conditions



Adiabatic mixing

Total mass balance: $m_1 + m_2 = m_3$

A balance: $m_1 x_{A1} + m_2 x_{A2} = m_3 x_{A3}$

Solve the two eq's:

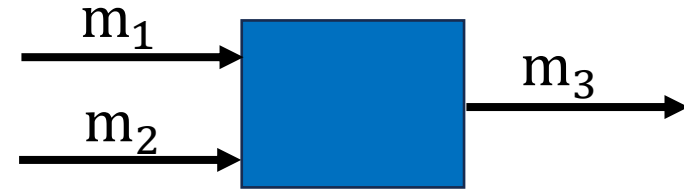
$$x_{A3} = \left(\frac{m_1 x_{A1} + m_2 x_{A2}}{m_1 + m_2} \right)$$

Energy balance: $\Delta H = m_3 \hat{H}_3 - m_1 \hat{H}_1 - m_2 \hat{H}_2 = 0$

$$m_1 (\hat{H}_3 - \hat{H}_1) - m_2 (\hat{H}_2 - \hat{H}_3) = 0$$

$$\frac{(\hat{H}_3 - \hat{H}_1)}{(x_{A3} - x_{A1})} = \frac{(\hat{H}_2 - \hat{H}_3)}{(x_{A2} - x_{A3})}$$

So three points must lie on a straight line: $x_{A3} = \left(\frac{m_1 x_{A1} + m_2 x_{A2}}{m_1 + m_2} \right)$



Adiabatic mixing

If two feed solutions of known masses and composition (m_i , x_i , $i = 1, 2$) are mixed adiabatically and a H-x chart is available;

- (i) Calculate x_3 for the product mixture by performing mass balance
- (ii) Draw a line connecting the points on the chart corresponding to the two feeds
- (iii) Find the enthalpy and temperature of the product mixture from the point on the connecting line for which $x = x_3$

Problem 1

An aqueous solution containing 85.0 wt% H₂SO₄ at 60 deg F (specific gravity 1.78) is diluted with pure liquid water at the same temperature. The feed solution volume is 350 mL. The mixing may be considered adiabatic, and the pressure is constant at 1 atm.

- a) The product solution is to contain 30.0 wt% H₂SO₄. Calculate the volume (mL) of water needed for the dilution, ideally using a single-dimensional equation.
- b) Use the enthalpy–concentration chart of Figure 8.5-1 to estimate the specific enthalpies (Btu/lbm) of the feed solution and the water. Then write an energy balance on this closed system constant pressure process and solve it for the specific enthalpy of the product solution. Finally, use Figure 8.5-1 to verify your calculated value of $\hat{H}_{product}$ and to estimate the product solution temperature.
- c) Use Figure 8.5-1 to estimate the maximum temperature that could be attained by mixing the feed solution with pure water and the concentration (wt% H₂SO₄) of the product solution.

Problem 2

50 mL of 100% H_2SO_4 at 25 deg C and 84.2 mL of liquid water at 15 deg C are mixed. The heat capacity of the product solution is 2.43 J/(g deg C). Estimate the maximum temperature attainable by the product solution and state the conditions under which this temperature would be attained, using heat of mixing data from Table B.11.

Problem 3

Suppose $m_A(\text{g})$ of species A [molecular weight M_A , heat capacity C_{pA} [J/(g. deg C)] at temperature T_{A0} (deg C) and $m_B(\text{g})$ of species B (M_B , C_{pB}) at temperature T_{B0} are mixed adiabatically. The heat of mixing of A and B at 25 deg C is $\Delta\hat{H}_m(r)$ (J/mol A in solution), where $r = (m_B/M_B)/(m_A/M_A)$. The heat capacity of the product solution is C_{ps} [J/(g. deg C)]. All heat capacities may be considered independent of temperature.

- Derive an expression for $T_{\max}$, the highest temperature attainable by the product solution, in terms of the other quantities defined. State the conditions that would have to be met for this temperature to be approached.
- Use your expression to estimate $T_{\max}$ for a process in which 100.0 g of sodium hydroxide at 25 deg C and 225.0 g of water at 40 deg C are combined to form a product solution with a heat capacity of 3.35 J/(g. deg C).

Problem 4

Aqueous sulfuric acid solutions containing 15.0 wt% H_2SO_4 and 80.0 wt% H_2SO_4 are mixed to form a 60.0 wt% product solution. The 15% solution was in a laboratory in which the temperature was 77 deg F. The 80% solution had just been taken from a storage cabinet in an air-conditioned stock room and was at a temperature of 60 deg F when the mixing occurred.

- a) The mass of the 15% solution is 2.30 lbm. What mass of 60% solution should be weighed out?
- b) Use Figure 8.5-1 to estimate the product solution temperature if the mixing is adiabatic.
- c) The product solution temperature eventually drops to (77 deg F). How much heat (Btu) is transferred from the solution to the laboratory air in this constant-pressure cooling process?